Labid Al Nahiyan

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RESEARCH INTERESTS

Large Language Models · Computer Vision · AI Safety & Alignment · Software Engineering

EDUCATION

Bangladesh University of Engineering and Technology (BUET) Feb 2020 – Mar 2025

B.Sc. in Computer Science and Engineering

Notable Courses: Machine Learning, Computer Security, Compiler Design, Database Systems, Software Engineering, Data Structures & Algorithms, OOP

RESEARCH EXPERIENCE

When Safety Blocks Sense: Measuring Semantic Confusion in LLM Refusals 2025

Co-Author | Preprint: arXiv:2512.01037

- Introduced **"Semantic Confusion,"** a failure mode where models incorrectly reject harmless paraphrases of intents they otherwise accept.
- Developed **ParaGuard**, a 10k-prompt corpus of controlled paraphrase clusters to benchmark model consistency.
- Proposed token-level metrics - Confusion Index, Rate, and Depth - using embeddings and perplexity signals to diagnose safety-alignment gaps.

Contextually Appropriate Loss Function for Dense Crowd Tracking 2024

Undergraduate Thesis | Supervisor: Prof. A. B. M. Alim Al Islam, BUET

- Proposed a size-aware, gradient-stabilized loss function to optimize the tracking of small, overlapping objects in dense crowd scenarios.
- Integrated the loss function into **YOLO-based models**; evaluated on JHU-CROWD++ and SCUT-HEAD benchmarks.
- Achieved superior mAP@50 and F1-scores compared to CIoU, DIoU, and GIoU baselines.

WORK EXPERIENCE

TigerIT Bangladesh Ltd. Aug 2025 – Present

Machine Learning Engineer (R&D)

- Reduced FNMR by **41%** ($0.0151 \rightarrow 0.0089$ @ FMR $\leq 10^{-2}$) on the **NIST MINEX III** benchmark via an optimized minutiae-based template generator.
- Engineered a custom loss function to address non-linear elastic distortions and rotational variances in fingerprint matching.

Technologies: Python, PyTorch, CUDA, Computer Vision

Intellesphere.AI May 2025 – Jul 2025

Full-Stack Developer

- Built responsive frontend components from UI/UX designs and integrated RESTful APIs for dynamic data rendering.
- Tested, debugged, and optimized the live-deployed web application to ensure full functionality.

Technologies: React.js, Express.js, Zustand

SELECTED PROJECTS

Shop Genie - AI E-commerce Assistant GitHub

SvelteKit, LLM, RAG

An AI-powered shopping assistant that enables natural language product search over a corpus of 10,000+ scraped product listings.

NerdHerd - Educational Platform GitHub

SvelteKit, Tailwind CSS, PostgreSQL

An all-in-one student platform featuring live classes, video calls, community forums, and quiz modules.

Bidirectional Autonomous Robot Website

Arduino, Hierarchical Q-Learning

An autonomous crawling robot featuring real-time obstacle avoidance and live policy adaptation via a custom Arduino setup.

Subset-of-C Compiler GitHub

C, Bison, Flex, YACC

Implemented lexical analysis, parsing, symbol table management, and assembly code generation for a C-language subset.

TECHNICAL SKILLS

Languages	Python, C/C++, Java, TypeScript, SQL
Frameworks	PyTorch, LangChain, LangGraph, React.js, Svelte, Express.js
Tools	Git, Docker, Hugging Face, Linux

HONORS & AWARDS

Champion - OpenAPI Hackathon, 11th National ICT Fest 2024

IUT, Bangladesh

7th Place - SEC Inter-University Junior Programming Contest 2022

Team: BUET Fast & Furious

REFERENCES

Prof. A. B. M. Alim Al Islam Bangladesh University of Engineering and Technology
Professor, Computer Science and Engineering alim_razi@cse.buet.ac.bd